

Craig T. Russell, PhD

Senior Machine Learning Scientist — Generative & Agentic Models for Biology

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Focus: Generative & agentic models for biology • LLM post-training • Out-of-distribution modelling • Virtual cells • Open to relocation (San Francisco)

PROFESSIONAL SUMMARY

ML Research Scientist building **generative models that generalise to unseen biology**, so that we can build the first virtual cell. Eight years applying machine learning to biological data, from light-sheet microscopy and TB-scale image archives to single-cell transcriptomics and knowledge graphs, with papers in *Nature Biotechnology*, *NeurIPS* and *Nature Methods*. I write the research code and the infrastructure: tools other labs use, agents grounded in real biological databases, and training and inference on cloud and HPC.

EXPERIENCE

Senior Machine Learning Scientist — Valence Labs @ Recursion Pharmaceuticals

London, UK • Oct 2024 – Present

- **Agentic biology:** Build **tool-use systems grounded in real biological databases** — constrained decoding over **Ensembl/HGNC/ChEMBL** vocabularies, tool-call verification against structured resources, retrieval grounding and verifier ensembles — producing explainable, multi-source gene–gene hypotheses.
- **TxPert** (*Nature Biotechnology*, 2026): Co-author on a state-of-the-art transcriptomic perturbation predictor conditioned on multiple biological knowledge graphs; owned benchmark-task design, data curation and **OOD ablations**.
- **LLM fine-tuning & reinforcement learning:** Post-train multi-modal LLMs over **knowledge graphs, literature and omics**; apply **RL** to align agent behaviour with verifiable biological objectives.
- **Generative out-of-distribution modelling:** Train **large diffusion transformers** to generate perturbational biology, targeting generalisation to **unseen** perturbations, cell types and their combinations rather than in-distribution interpolation.
- **Gene expression + bioimage data:** Model **single-cell and bulk RNA-seq** jointly with **high-content phenotypic imaging**, so each modality constrains the other where one alone leaves the prediction under-determined.
- **Active learning:** Design acquisition strategies that choose the next experiment under a fixed budget, closing the loop between model uncertainty and wet-lab capacity.
- **Research engineering lead, ~30-person research team:** Set technical direction and engineering standards across a multidisciplinary ML and biology group; **scale models to the cloud** for multi-GPU training and distributed inference; foster collaborative coding — code review, reproducibility, shared tooling — to carry research prototypes into production.
- **Virtual screening:** Evaluation strategy and error analysis for proteome-scale binding-affinity screening.
- **Community:** Organiser, Virtual Cell Journal Club, bridging ML and wet-lab teams.

Senior Research Associate & AI Engineering Lead — EMBL-EBI (Uhlmann Group & Bio-Image Archive)

Cambridge, UK • Dec 2022 – Oct 2024

- **Team leadership:** Supervised **6 PhD students**; established coding standards, CI and peer review across the lab.
- **Self-supervised representation learning:** Created [bioimage_embed](#) and co-authored **ShapeEmbed** (*NeurIPS* 2025) — contour and morphology representations learned without labels, both shipped as **production Python used by other labs**.
- **Cloud compute for biologists:** First author on **bia-binder** (*Bioinformatics* 2025) — web-native cloud compute alongside the Bio-Image Archive.
- **MLOps:** Designed scalable pipelines processing **TB-scale microscopy** across HPC and cloud; containerised workflows, automated experiment tracking.

Part-time and consulting roles held alongside EMBL-EBI, 2022 – 2024:

AI/ML Founding Engineer (part-time) — Amun AI AB

Stockholm, Sweden • 2022 – 2024

- Built a **GKE/Kubernetes** model-serving platform with **NVIDIA Triton/KServe**; supported **100+ models** for **30+ daily users** with auth, monitoring and autoscaling.

AI/ML Engineering Consultant (part-time) — DeepMirror

Cambridge & London, UK • 2022 – 2024

- **MouseMindMapper:** Shipped a brain-histology segmentation product generating **£50k annual revenue**; owned it end to end — data, training, packaging and docs.
- Wrote a high-performance **C++ cheminformatics fingerprinting** library for production.

Data Scientist — Bzarma Group, EMBL-EBI

Cambridge, UK • Dec 2019 – Dec 2023

- Co-authored the successful **AI4LIFE €5M** grant (federated bioimage AI infrastructure); contributed to platform architecture and model-sharing strategy.
- Drove **large-scale AI microscopy** analyses in the Image Data Resource; collaborated with **Google Cloud** on representation learning.
- Taught annual deep-learning courses to **40+ researchers** (PhD to PI).

Software Engineer (COVID-19 Response) — European Nucleotide Archive, EMBL-EBI

Cambridge, UK • Mar 2020 – Sept 2020

- Built CI/CD for the **COVID-19 Data Portal** enabling **daily global data updates**; scaled NGS alignment and **Nextflow/Kubernetes** ETL for surging data volumes.

Computational Microscopist — National Physical Laboratory

London, UK • 2018 – Dec 2019

- Developed novel **3D organoid segmentation** methods for cancer research; consulted for MSquared on advanced imaging.

EDUCATION

PhD, Engineering — University of Cambridge • 2014 – 2018 (EPSRC PES-CDT)

Thesis: “Light-sheet microscopy for tracking particles in large specimens”

- Designed and built a novel light-sheet microscope with automated acquisition; algorithms for particle tracking, signal optimisation and micrometre-scale tomography. Supervised 2× MRes and 1× BSc students.

MRes, Photonics — University of Cambridge & UCL • 2013 – 2014 • structured-illumination microscopy reconstruction

MSci, Physics (First-Class Honours) — Nottingham Trent University • 2009 – 2013 • top physics graduate

SELECTED PUBLICATIONS

1. **TxPert** — out-of-distribution transcriptomic perturbation prediction over biological knowledge graphs. *Nature Biotechnology* (2026) • co-author. [doi:10.1038/s41587-026-03113-4](https://doi.org/10.1038/s41587-026-03113-4)
2. **ShapeEmbed** — self-supervised learning of 2D contour representations. *NeurIPS* (2025) • second author. [proceedings](#)
3. **bia-binder** — web-native cloud compute for bioimage analysis. *Bioinformatics* (2025) • first author. [doi:10.1093/bioinformatics/btaf412](https://doi.org/10.1093/bioinformatics/btaf412)
4. **MIFA** — metadata and accessibility standards for reusable AI training datasets in bioimaging. *Nature Methods* (2025) • co-author. [doi:10.1038/s41592-025-02835-8](https://doi.org/10.1038/s41592-025-02835-8)
5. **DL4MicEverywhere** — reproducible, containerised deep learning for microscopy. *Nature Methods* (2024) • co-author. [doi:10.1038/s41592-024-02295-6](https://doi.org/10.1038/s41592-024-02295-6)
6. **CIR4MICS** — synthetic ground truth for benchmarking image-analysis methods. *Bioinformatics* (2023) • co-author. [doi:10.1093/bioinformatics/btad587](https://doi.org/10.1093/bioinformatics/btad587)
7. **The COVID-19 Data Portal** — rapid open-access data sharing for SARS-CoV-2 research. *Nucleic Acids Research* 49(W1) (2021) • co-author. [doi:10.1093/nar/gkab417](https://doi.org/10.1093/nar/gkab417)
8. **mmSIM** — open toolbox for structured illumination microscopy. *Phil. Trans. R. Soc. A* (2021) • first author. [doi:10.1098/rsta.2020.0353](https://doi.org/10.1098/rsta.2020.0353)
9. **Frame-localisation OPT** — reconstruction for optical projection tomography. *Scientific Reports* (2021) • first author. [doi:10.1038/s41598-021-83454-z](https://doi.org/10.1038/s41598-021-83454-z)

Full list on [Google Scholar](#).

PATENTS

- **Virtual Cell Foundation Model** • Patent pending • 2024 (Recursion)
- **TxPert: Transcriptomic Perturbation Prediction** • Patent pending • 2024 (Recursion)

OPEN SOURCE

[bioimage_embed](#) (author) • contributions to **napari**, **BiolImage Model Zoo**, **Hypha Platform**, **DL4MicEverywhere**, **BIA Binder**, **COVID Workflow Manager**

SKILLS

ML & AI: Diffusion transformers, generative modelling, foundation-model fine-tuning & post-training, reinforcement learning, agentic / tool-use systems, constrained decoding & retrieval grounding, active learning, contrastive/self-supervised learning, OOD & uncertainty, evaluation/benchmark design

Frameworks: PyTorch, Lightning, Hugging Face, Pyro, TensorFlow, scikit-learn

Bio & data: Single-cell & bulk RNA-seq, high-content & phenotypic imaging, histopathology, GNNs, knowledge graphs; UniProt, PDB, Ensembl, NCBI, ChEMBL

Languages: Python (primary), R, C++, Rust, MATLAB, Java

Compute: Multi-GPU (A100/V100), CUDA, distributed training, SLURM, HPC, GCP/AWS

MLOps/Infra: Kubernetes, Docker, NVIDIA Triton, KServe, MLflow, CI/CD, Terraform; Nextflow, Snakemake, Airflow

GRANTS, TEACHING & SERVICE

- **Grants:** AI4LIFE (2022, €5M EU Horizon, co-author) • EPSRC CDT Studentship (2013–2018, £120k) • Nuffield Research Bursary (2012)
- **Course lead:** Deep Learning for Bioimage Analysis (2019–2023), 40+ participants/year
- **Supervision:** 6 PhD students (AI & spatial biology) + 3 project students
- **Peer review:** Nature Methods, Scientific Reports, Journal of Microscopy, ISBI (2022, 2023), ICASSP (2024)
- **Talks:** FOM (2018, 2022, 2023), MMC (2018, 2022), CBIAS (2023)